

Online Resource 1: Supporting analyses for size-dependent demography, disturbance, and transition dynamics of *Acropora palmata* across the Caribbean

Adrian C. Stier; A. Raine Detmer; Jameal F. Samhouri; Darcy Bradley; Aldo Croquer; Rita I. Sellares-Blasco

Journal: *Coral Reefs*

Corresponding author: Adrian C. Stier, Department of Ecology, Evolution, and Marine Biology and Marine Science Institute, University of California, Santa Barbara, CA 93106, USA.

Corresponding author email: astier@ucsb.edu.

Contents

1. Studies contributing to the synthesis (Table S1)
2. Study-window disturbance summary (Table S2)
3. Growth-size diagnostics (Fig. S1)
4. Population origin is confounded with study identity (Fig. S2)
5. Shrinkage and retrogression (Fig. S3)
6. Disturbance and size-dependent survival (Fig. S4)
7. Restoration subtype sensitivity (Fig. S5)
8. Florida-scale heatwave scenarios (Fig. S6)

1. Studies contributing to the synthesis

Table S1. Eighteen studies contributed data; seventeen entered the survival synthesis. Tier 1 studies supplied individual colony records; Tier 2 studies supplied study-level summaries. N is the effective annualized denominator in the survival synthesis after interval filtering and aggregation, summed across a study’s effects; survival is the N -weighted mean of annualized effect-level survival and is shown for orientation only (Fig. 3 gives random-effects estimates). Roles: S, survival synthesis; V, size-dependent vital-rate models; M, transition matrix (natural-colony survival and growth). Fragmentation in the matrix came only from Vardi (2011), whose summaries also entered the survival synthesis.

Study	Tier	Region(s)	Population type	Effects	N	Survival (%)	Roles
NOAA monitoring	1	Florida Keys, Curaçao, Navassa	Natural	3	3,874	87.4	S, V, M
Neely et al. (2022)	1	Florida Keys	Natural	1	2,191	70.2	S, V, M
Pausch et al. (2018)	1	Florida Keys	Restoration	1	968	57.4	S, V
Kuffner et al. (2020)	1	Florida Keys	Restoration	1	53	81.1	S, V
USGS outplant experiment (unpublished)	1	US Virgin Islands	Restoration	1	46	65.5	S, V
FUNDEMAR fragments (unpublished)	1	Dominican Republic	Restoration	1	44	86.6	S, V
Mendoza Quiroz et al. (2023)	1	Mexican Caribbean	Restoration (sexual outplants)	–	–	–	V
Vardi (2011)	2	Jamaica, Puerto Rico, Virgin Gorda	Natural	3	245	86.9	S, M
Bruckner and Bruckner (2001)	2	Puerto Rico	Restoration	1	105	90.5	S
Ortiz Prosper (2005)	2	Puerto Rico	Restoration	1	207	78.7	S
Forrester et al. (2014)	2	British Virgin Islands	Restoration	1	257	53.3	S
Rosales et al. (2024)	2	Florida Keys	Restoration	1	58	79.3	S
Maurer et al. (2022)	2	Bahamas	Restoration	1	24	95.8	S
Williams and Miller (2010)	2	Florida Keys	Restoration	1	18	77.8	S
Garrison and Ward (2008)	2	US Virgin Islands	Natural and restoration	2	75	69.3	S
Rogers and Muller (2012)	2	US Virgin Islands	Natural	1	69	94.2	S
Ramos Romero et al. (2025)	2	Cuba	Restoration	1	200	70.5	S
Rogers et al. (1982)	2	US Virgin Islands	Natural	1	173	48.0	S
Total				22	8,607		

2. Study-window disturbance summary

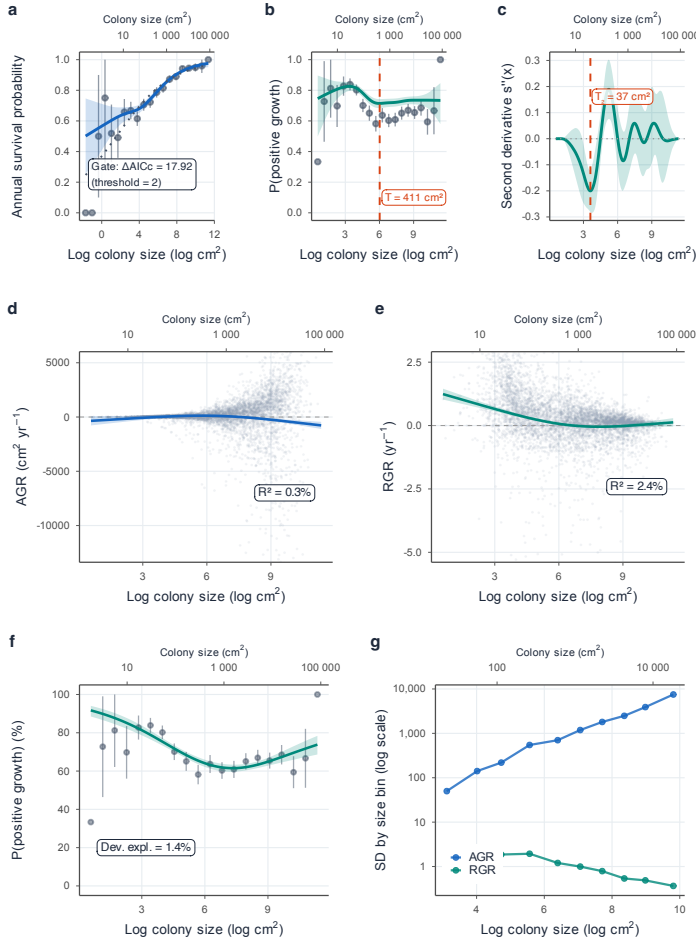
Table S2. Curated disturbance overlap was pervasive across the demographic records. The study-window summary links 1,072 prepared survival intervals from the seven Tier 1 studies to the curated disturbance timeline. Only 53 intervals (4.9%) overlapped no curated event. “Baseline exclusion” identifies the 297 intervals removed only in the acute-event sensitivity analysis; it does not mean that other intervals were undisturbed.

Study	Intervals	No overlap	One event	Two or more events	Baseline exclusion	Any overlap (%)
NOAA monitoring	842	50	59	733	228	94.1
Neely et al. (2022)	98	3	0	95	10	96.9
Pausch et al. (2018)	70	0	0	70	46	100.0
USGS outplant experiment	27	0	15	12	12	100.0
Kuffner et al. (2020)	27	0	0	27	0	100.0
Mendoza Quiroz et al. (2023)	7	0	0	7	0	100.0
FUNDEMAR fragments	1	0	0	1	1	100.0
Total	1,072	53	74	945	297	95.1

The linkage produced zero field mismatches. Full disturbance chronology, interval-level assignments, and extended diagnostics are available in the project repository.

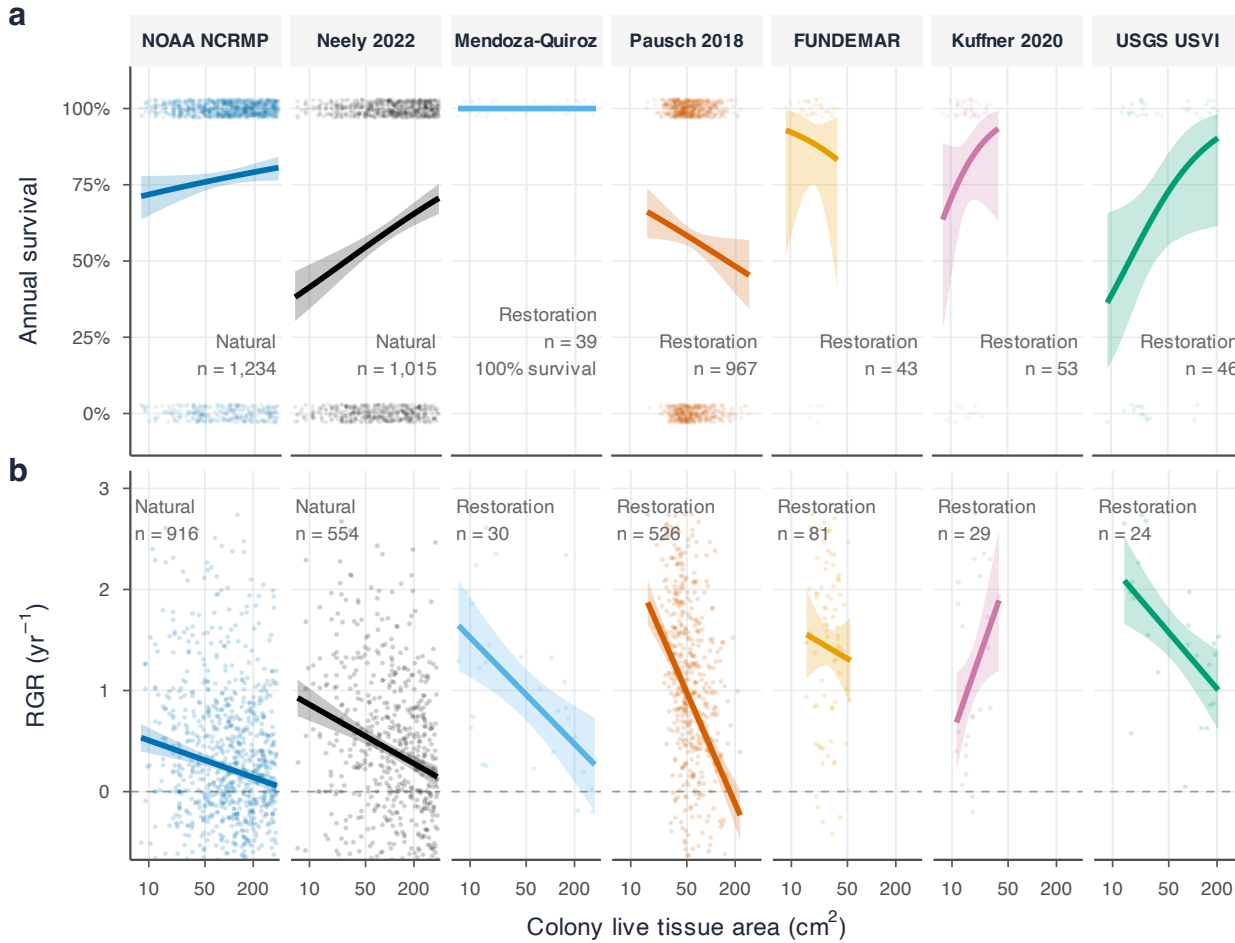
3. Growth-size diagnostics

Fig. S1. Survival and growth change with colony size at different points in ontogeny. (a) Annual survival against log live planar area; the label gives the improvement of the GAM over a linear model ($\Delta\text{AICc} = 17.9$; the pre-specified gate for reporting a threshold was 2). (b) Probability of positive growth; the dashed line marks the supported threshold (411 cm^2). (c) Second derivative of the relative-growth-rate GAM; the dashed line marks the steepest change in curvature (37 cm^2). (d, e) Absolute growth rate (AGR, $\text{cm}^2 \text{yr}^{-1}$) and relative growth rate (RGR, yr^{-1}) against colony size. (f) Probability of positive growth fitted with a lower-dimension smooth ($k = 3$), which captures the broad U-shaped trend but not the dome detected in panel b. (g) Standard deviation of AGR and RGR within size bins on a log scale: AGR variance rose by more than two orders of magnitude across the size range, whereas RGR variance declined. Points are binned observations, lines are GAM fits, and ribbons or error bars are 95% confidence intervals. Analyses exclude the upper and lower 0.5% of relative-growth values to limit leverage by extreme proportional changes and exclude high-variance Navassa records.



4. Population origin and study identity

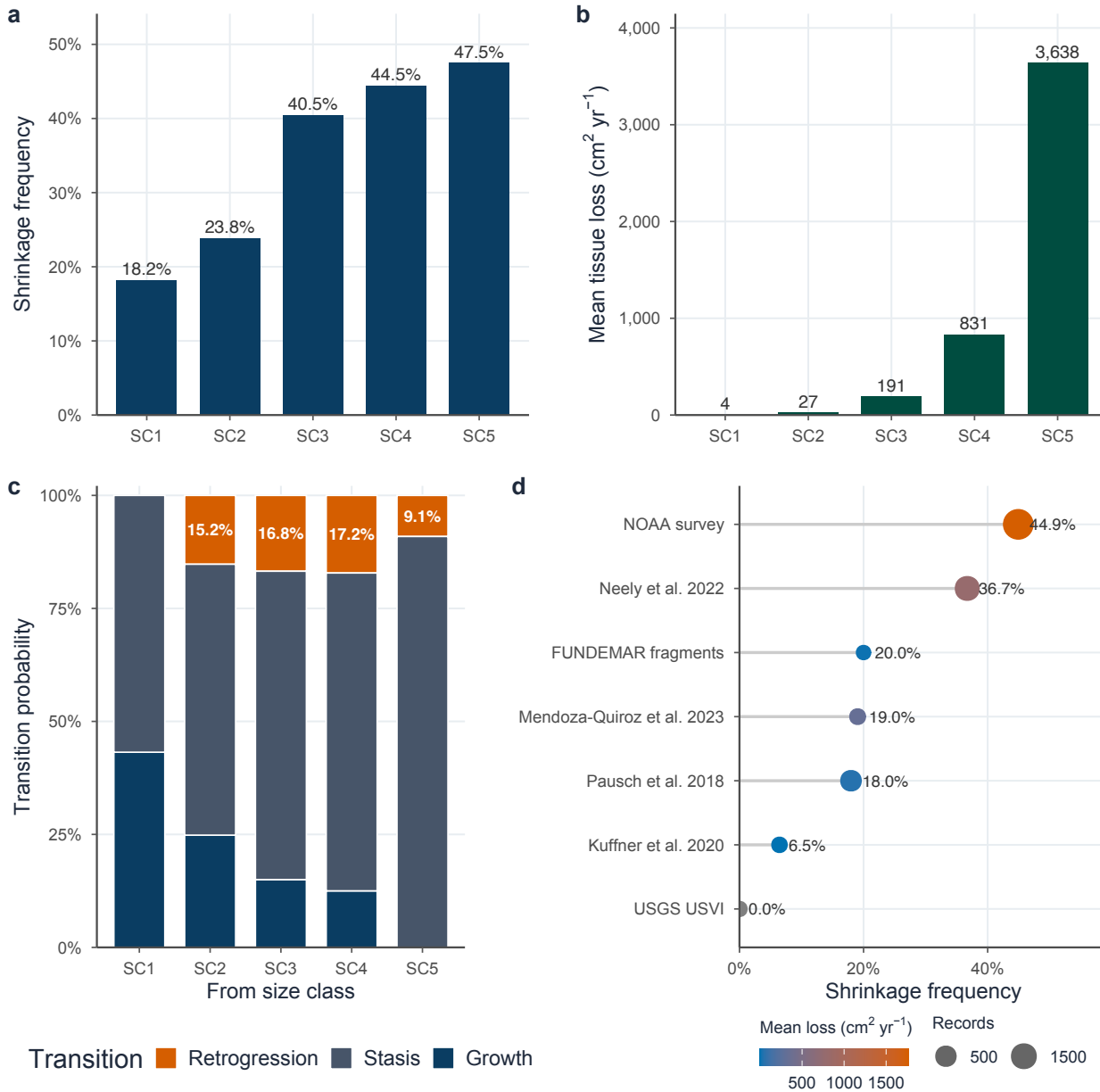
Fig. S2. Natural-colony and restoration-fragment comparisons are confounded by study identity and a narrow shared size range. Each facet shows one Tier 1 study within the size range shared by natural and restoration records. (a) Annual survival and (b) relative growth rate (yr^{-1}) against initial live tissue area (cm^2). Points are observations, coloured lines are study-specific binomial GLM (a) or linear model (b) fits on log size, shaded bands are 95% confidence intervals, and labels give population type and sample size. The shared range (11–202 cm^2) is small relative to the full 1–15,000 cm^2 range, and within it survival slopes differed in sign among studies, so this comparison is descriptive rather than a causal estimate of restoration origin.



Study identity confounded with population type. Shared size range: 11–202 cm^2 of ~1–15,000 cm^2 .

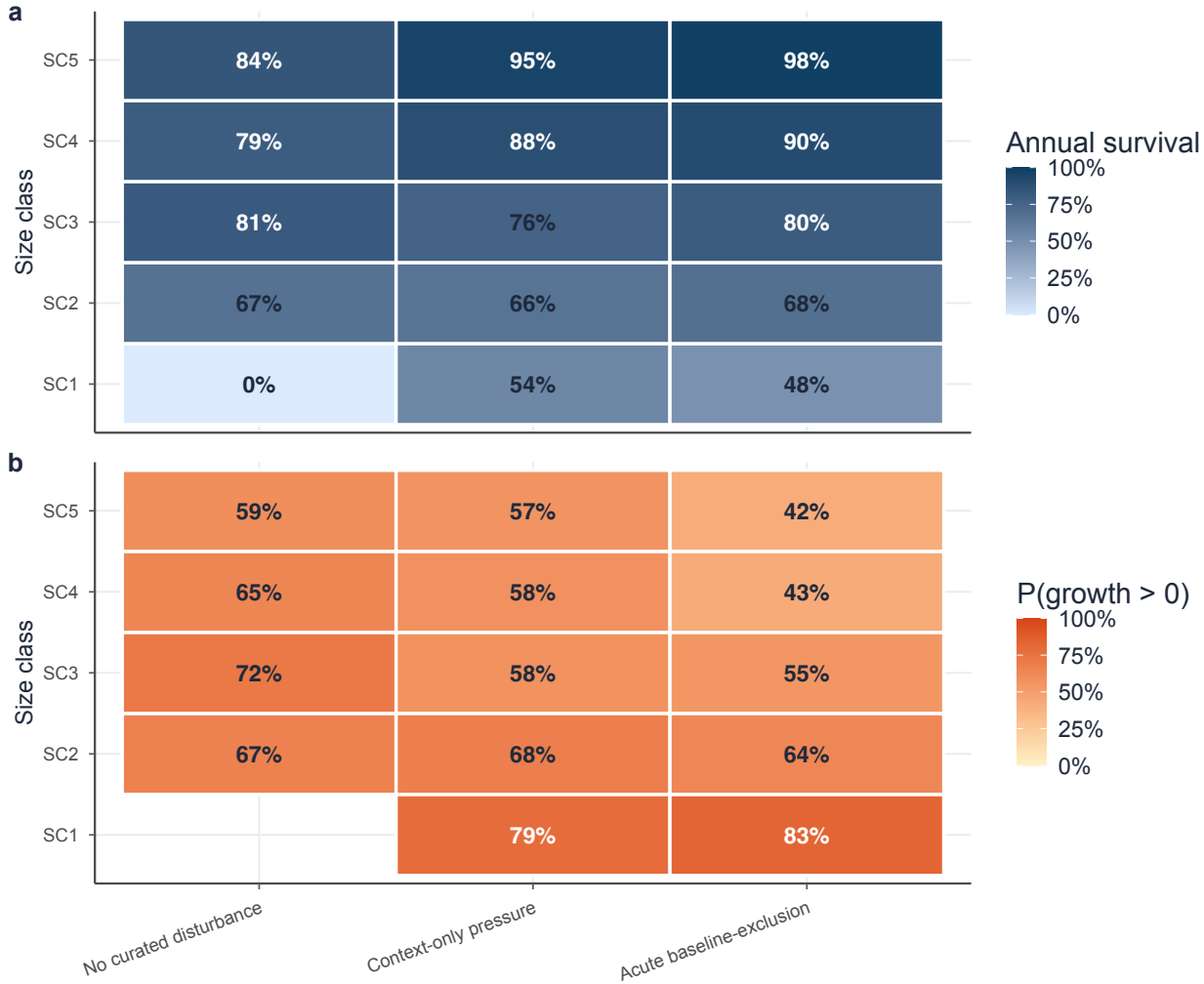
5. Shrinkage and retrogression

Fig. S3. Large colonies commonly survived while losing tissue, and every non-recruit class moved backward through the size structure. (a) Shrinkage frequency among surviving colonies by size class, rising from 18.2% in SC1 to 47.5% in SC5. (b) Mean annual tissue loss among shrinking colonies ($\text{cm}^2 \text{yr}^{-1}$). (c) Observed annual transition probabilities (growth, stasis, retrogression) by initial size class; labels give retrogression probabilities. (d) Shrinkage frequency by study; point colour gives mean tissue loss and point size the number of records. The analysis uses matrix-compatible records of surviving colonies with paired size observations.



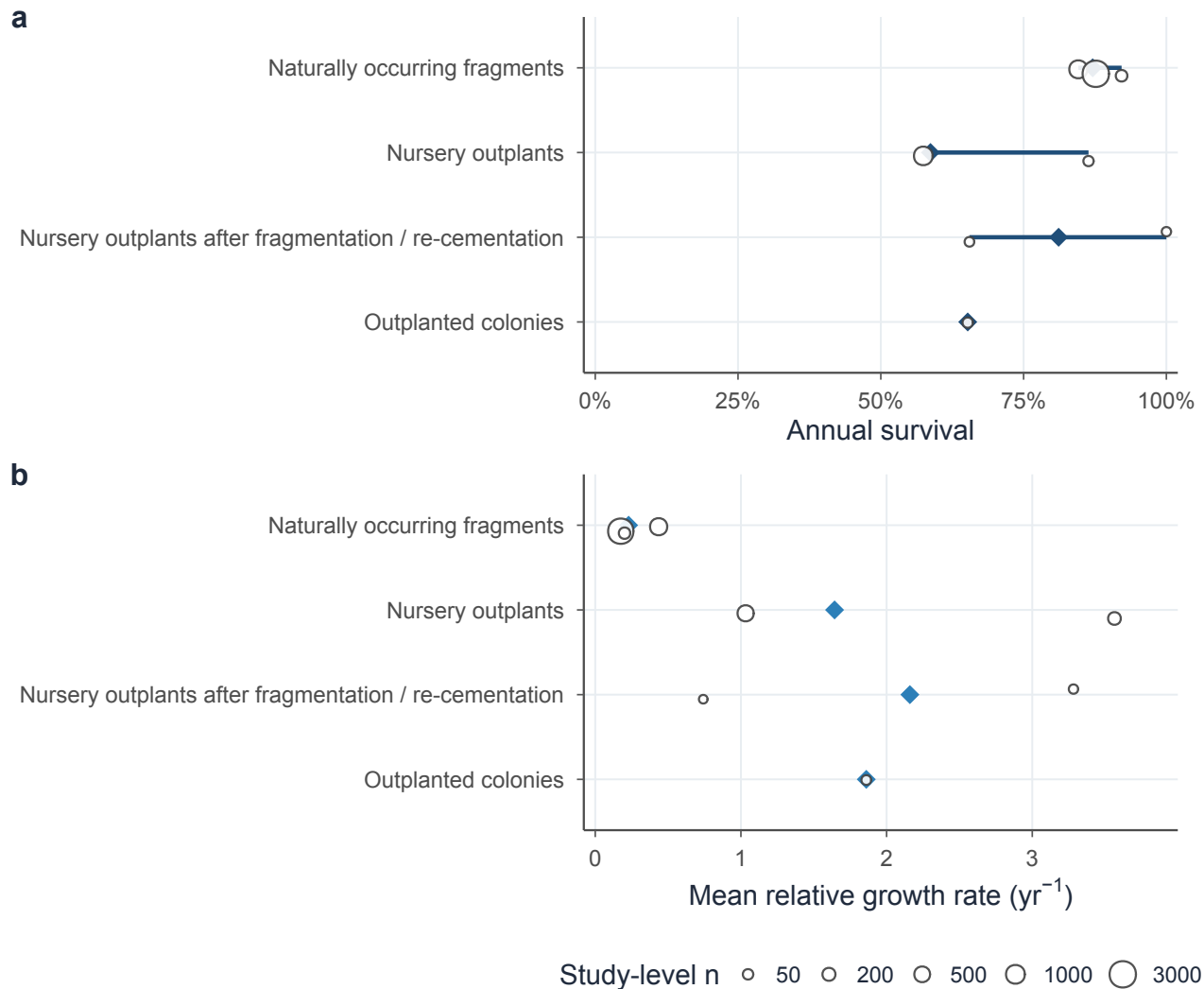
6. Disturbance and size-dependent survival

Fig. S4. The relationship between size and survival differed among curated disturbance states, but not as a simple disturbance penalty. Tiles give observed (a) annual survival and (b) probability of positive growth by size class and disturbance state for natural colonies ($n = 6,141$ survival records). Tile labels are percentages. The size-by-state interaction was supported for survival (likelihood-ratio test, $p < 0.001$) and weaker for positive growth ($p = 0.010$). Large colonies survived better in intervals flagged for acute events (SC5, 98.0%, $n = 409$) than in intervals with no curated event (SC5, 83.9%, $n = 193$). The no-disturbance category contained only 353 records, including a single SC1 colony, so its SC1 tile is not interpretable. Disturbance states are observational and confounded with study, site, and survey period.



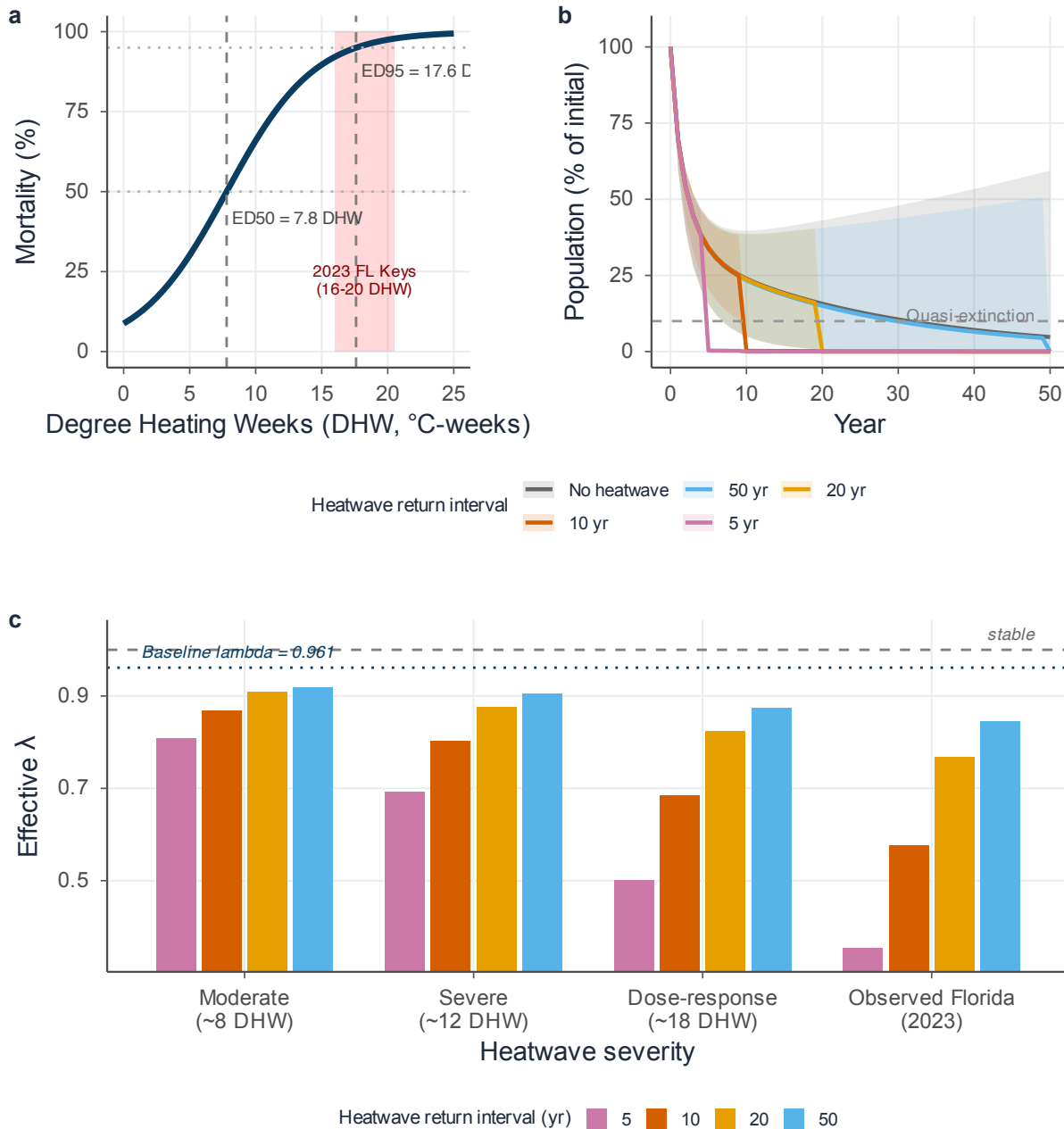
7. Restoration subtype sensitivity

Fig. S5. Apparent restoration performance depends on the biological material grouped under that label. (a) Annual survival and (b) mean relative growth rate (yr^{-1}) for naturally occurring fragments (3,968 records, one study), nursery outplants (1,012 records, two studies), nursery outplants re-cemented after fragmentation (53 records, one study), and outplanted colonies (46 records, one study). Open circles are study summaries, with symbol size proportional to records; filled diamonds are record-weighted subtype means, and horizontal lines span the study summaries. These contrasts are descriptive because subtype, study, site, handling, and sample size are not independently balanced.



8. Florida-scale heatwave scenarios

Fig. S6. Recurrence of Florida-scale heatwave mortality drives effective lambda far below one in the recruitment-free transition subsystem. (a) Logistic mortality curve parameterized from the *A. palmata* DHW thresholds reported by Manzello et al. (2025) (ED50 = 7.8 DHW; ED95 = 17.6 DHW); the shaded range marks the 16–20 DHW exposure in the Florida Keys in 2023. (b) Median 50-year trajectories and 95% simulation intervals without heatwaves and for the observed 2023 Florida endpoint (97.8–100% mortality) at four return intervals; the dashed line marks the quasi-extinction threshold of 10% of initial abundance. (c) Effective lambda across dose-response and observed-endpoint scenarios; bars are geometric means from the bootstrap-based simulations, and the dotted line is the unperturbed deterministic lambda (0.961). Without heatwaves, the simulations had an effective lambda of 0.933 and a 70.8% probability of quasi-extinction within 50 years; every observed-endpoint scenario reached quasi-extinction in all simulations. The 5-, 10-, 20-, and 50-year intervals span frequent to rare recurrence and are sensitivity values, not estimated return periods. These are Florida-like event stress tests, not Caribbean-wide forecasts.



References

- Bruckner AW, Bruckner RJ (2001) Condition of restored *Acropora palmata* fragments off Mona Island, Puerto Rico, 2 years after the Fortuna Reefer ship grounding. *Coral Reefs* 20:235–243. <https://doi.org/10.1007/s003380100164>
- Forrester GE, Ferguson MA, O’Connell-Rodwell CE, Jarecki LL (2014) Long-term survival and colony growth of *Acropora palmata* fragments transplanted by volunteers for restoration. *Aquatic Conservation: Marine and Freshwater Ecosystems* 24:81–91. <https://doi.org/10.1002/aqc.2374>
- Garrison V, Ward G (2008) Storm-generated coral fragments – A viable source of transplants for reef rehabilitation. *Biological Conservation* 141:3089–3100. <https://doi.org/10.1016/j.biocon.2008.09.020>
- Kuffner IB, Stathakopoulos A, Toth LT, Bartlett LA (2020) Reestablishing a stepping-stone population of the threatened elkhorn coral *Acropora palmata* to aid regional recovery. *Endangered Species Research* 43:461–473. <https://doi.org/10.3354/esr01083>
- Manzello DP, Cuning R, Karp RF, et al (2025) Heat-driven functional extinction of Caribbean *Acropora* corals from Florida’s Coral Reef. *Science* 390:361–366. <https://doi.org/10.1126/science.adx7825>
- Maurer L, Puishys L, Ho NKP, et al (2022) *Acropora cervicornis* and *Acropora palmata* cultured on a low maintenance line nursery design in The Bahamas. *PLoS ONE* 17:e0267034. <https://doi.org/10.1371/journal.pone.0267034>
- Mendoza Quiroz S, Beltrán-Torres AU, Grosso-Becerra MV, et al (2023) Long-term survival, growth, and reproduction of *Acropora palmata* sexual recruits outplanted onto Mexican Caribbean reefs. *PeerJ* 11:e15813. <https://doi.org/10.7717/peerj.15813>
- Neely KL, Macaulay KA, Lunz KS (2022) Population trajectory and stressors of *Acropora palmata* sites in the Florida Keys. *Frontiers in Marine Science* 9:978785. <https://doi.org/10.3389/fmars.2022.978785>
- Ortiz Prosper AL (2005) Population dynamics of hurricane-generated fragments of elkhorn coral *Acropora palmata* (Lamarck, 1816). {PhD} dissertation, University of Puerto Rico, Mayagüez
- Pausch RE, Williams DE, Miller MW (2018) Impacts of fragment genotype, habitat, and size on outplanted elkhorn coral success under thermal stress. *Marine Ecology Progress Series* 592:109–117. <https://doi.org/10.3354/meps12488>
- Ramos Romero A, González-Díaz P, Aguilera Pérez G, Banaszak AT (2025) Does reef crest zone selection influence *Acropora palmata* (Lamarck, 1816) fragment survival and growth? *PeerJ* 13:e20303. <https://doi.org/10.7717/peerj.20303>
- Rogers CS, Muller EM (2012) Bleaching, disease and recovery in the threatened scleractinian coral *Acropora palmata* in St. John, US Virgin Islands: 2003–2010. *Coral Reefs* 31:807–819. <https://doi.org/10.1007/s00338-012-0898-8>
- Rogers CS, Suchanek TH, Pecora FA (1982) Effects of Hurricanes David and Frederic (1979) on shallow *Acropora palmata* reef communities: St. Croix, U.S. Virgin Islands. *Bulletin of Marine Science* 32:532–548
- Rosales SM, Young BD, Bright AJ, et al (2024) Reef site and habitat influence effectiveness of *Acropora palmata* restoration and its microbiome in the Florida Keys. *Communications Earth & Environment* 5:689. <https://doi.org/10.1038/s43247-024-01816-7>
- Vardi T (2011) The threatened Atlantic elkhorn coral, *Acropora palmata*: Population dynamics and their policy implications. {PhD} dissertation, University of California, San Diego
- Williams DE, Miller MW (2010) Stabilization of fragments to enhance asexual recruitment in *Acropora palmata*, a threatened Caribbean coral. *Restoration Ecology* 18:446–451. <https://doi.org/10.1111/j.1526-100X.2009.00579.x>